

Name: Sarthak Subhash Dumbare  
Division: B PRN: 202201040038

Q. 1.

Similarities:

- 1) Task Management: All major OS handle multitasking by managing processes & threads, allocating CPU time & maintaining system responsiveness.
- 2) File Systems: Each OS manages files & directories, ensuring data is stored, retrieved, and organized. They support various file systems (e.g., NTFS for windows, HFS+ for Mac OS & ext4 for linux).
- 3) Device management: They provide drivers & to enable communication between OS & hardware devices.
- 4) User interfaces: OSs provide UIs, either command line or graphical, to interact with the system.



## Differences:

### 1) Kernel Architecture

#### Monolithic

Monolithic kernels: examples include linux; they manage all system functions within a single large kernel which enhances performance but is complex to maintain.

Microkernels: Examples are Minix & QNX.

They keep core functions minimal and run most services in user space.

### 2) System Compatibility:

i) Windows: often favoured in gaming & enterprise environment due to broad application support.

ii) Linux: Valued for flexibility, open-source nature & extensive customization.

### 3) Package management:

Windows: uses installers & updates via tools like Windows update.

Linux: Utilizes package managers (APT, RPM) for software installation & updates.



• Implications:

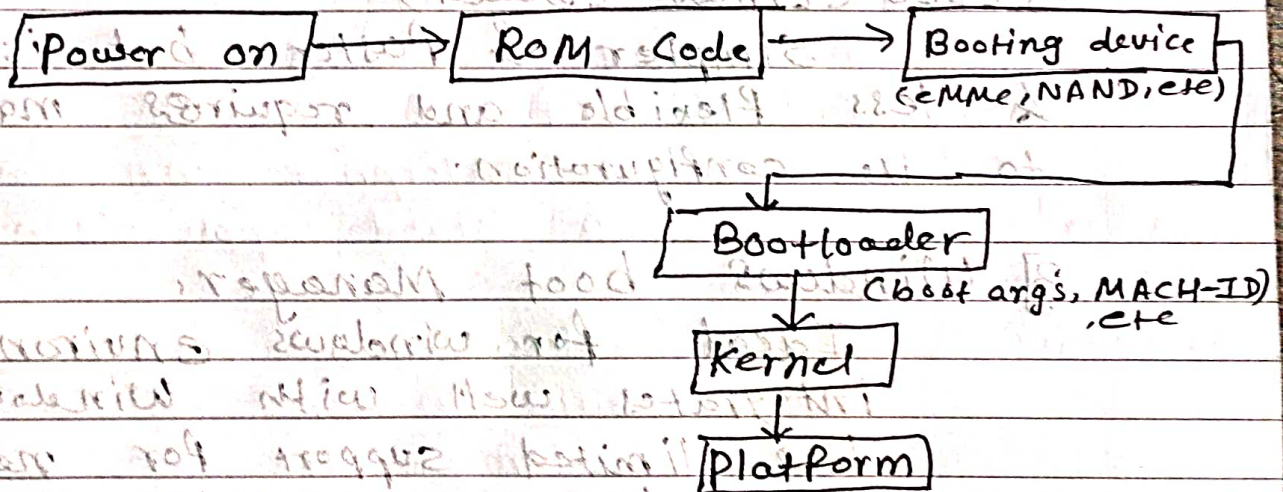
For users: The choice of OS affects software compatibility, system performance and overall user experience.

eg. Windows will give broader range of software's.

Linux will give more control & customization.

For Developers: The OS choice will impact development tools, deployment environments, and target audiences.

Q.2 → Working of Bootloader



- Functions:
- ① The main function of bootloader is to load the operating system.
  - ② Bootloaders are used to launch the application programs.
  - ③ Bootloaders are also used to expand or add missing functions firmware.



## Comparison of some common bootloaders

### 1] GRUB (GRand Unified Bootloader)

Advantages: i) Flexibility: supports multiple OSs & file systems.

ii) Configurations: Allows for complex configurations.

Disadvantages:

i) Complexity: mistakes in configurations can lead to boot issues.

ii) Performance: Can have slight performance overhead.

### 2] LILO (Linux Loader)

Simpler & faster but is outdated & less flexible and requires manual updates to its configuration.

### 3] Windows boot Manager.

Ideal for windows environments & integrates well with windows, but

has limited support for non-windows OS & less flexibility in boot management.